Master Electronic Learning Document

Everything You Need To Know about Electronics (The Basics)

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Document Description

This document is SSI's comprehensive list of resources for you to learn everything you need to learn about electronics. From voltage and current, to reading datasheets and using KiCad, if you're ever feeling lost, this is the first place you should check. The following list is written in the exact order it is expected to be read. Figure out your skill level, and start reading!

Tips for Starting

If this is your first time learning about electronics, consider reading through each topic of the **Electrical Theory Basics** section and watching the video corresponding to that topic. This will help you be able to understand the video the first time, without having to pause and catch up, in addition to clarifying/solidifying your understanding.

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Kale's Favourite Resources

Comprehensive, browsable guides/text resources to electronics. These will be your best friends in the world.

- **https://www.electronics-tutorials.ws/**
- **https://www.allaboutcircuits.com/textbook/**

Quick Learning for EE Beginners!

- Familiarize yourself with the basics of
  - **Basics of Electricity** (Video Guide / Text Guide)
  - **Current** (Video Guide / Text Guide)
  - **Voltage** (Video Guide / Text Guide)
  - **Power** (Video Guide (jump to 2:25) / Text Guide)
  - **Ohm's Law** (Video Guide / Text Guide)
  - **Kirchoff's Laws** (Video Guide / Text Guide)
  - **Series / Parallel Circuits** (Video Guide / Text Guide)
- Understand the following components:
  - **Inductors** (Video Guide / Text Guide)
  - **Capacitors** (Video Guide / Text Guide)
  - **Diodes** (Video Guide / Text Guide)
  - **Reading Electrical Schematics** (Video Guide / Text Guide)
  - **MOSFETs** (Video Guide / Text Guide) [optional, recommended]
    - **Metal-Oxide-Semiconductor Field-Effect Transistors**
  - **BJTs** (Video Guide / Text Guide) [optional, recommended]
- **Learn to use Kicad** (Getting Started / Schematics / PCB Layout)

Theory & Electronic Basics

Electrical Theory Basics

- What Are Electric Circuits? | Basic Concepts Of Electricity | Electronics Textbook
- Voltage and Current | Basic Concepts Of Electricity | Electronics Textbook
- Resistance | Basic Concepts Of Electricity | Electronics Textbook
- Voltage and Current in a Practical Circuit | Basic Concepts Of Electricity | Electronics Textbook
- How Voltage, Current, and Resistance Relate | Ohm's Law | Electronics Textbook
- Understanding Current-Voltage Curves - Technical Articles
- Electric Power - SparkFun Learn
- Guide to Passive Components - Peerless Electronics
- Passive Components in Electrical Circuits - GeeksforGeeks
- Introduction to Diodes And Rectifiers

Video Guides

- How ELECTRICITY works - working principle
- Electrical Current Explained - AC DC, fuses, circuit breakers, multimeter, GFCI, ampere
- Voltage Explained - What is Voltage? Basic electricity potential difference
- Ohms Law Explained - The basics circuit theory
- DC Series Circuits Explained
- DC parallel circuits explained - The basics how parallel circuits work working principle
- How Batteries Work
- Kirchhoff's Laws - Electricity - A-level Physics
- Electric Power Formula
- Intro to Inductors
- Diodes Explained - The basics how diodes work working principle pn j...
- Transistors Explained - How transistors work

Intro to Microcontrollers/Single Board Computers

- Introduction to Microcontrollers
- Getting started with Raspberry Pi Pico
- RP2350 - Wikipedia
- Getting started - Raspberry Pi Documentation

Intro to Sensors

- Introduction to Electronic Sensors
- Color Sensors Guide: Types, Calibration & Design Tips
- Overview of Photoelectric Sensors | OMRON Industrial Automation

Intro to Actuators

- A Brief Guide to Electric Actuators
- Introduction to Electric Motors

Intro to Communication Protocols

SPI Protocol

- Serial Peripheral Interface (SPI) - SparkFun Learn

I2C Protocol

- I2C - SparkFun Learn

UART Protocol

- Understanding UART | Rohde & Schwarz

Intro to Solar Cells + MPPTs

- Maximum power point tracking - Wikipedia
- Introduction to Solar Cells and Their Types
- Maximum Power Point Tracking (MPPT) algorithms
- Implementing a Simple Maximum Power Point Tracking (MPPT) Algorithm

Intro to Batteries

- What is a Battery? - SparkFun Learn
- Introduction to Battery Management Systems - Technical Articles

Intro to DC Power Regulation

- Regulator Topologies for Battery-Powered Systems | Analog Devices
    - Covers Buck/Boost etc.
- Buck Converters (Step-Down Converter)

Application, Assembly, and Testing

Intro to Breadboard Prototyping

- How to Use a Breadboard - SparkFun Learn
- Wire Gauges | Wires and Connections | Adafruit Learning System

Intro to Debugging & Testing Electronic Designs (Instrumentation)

- How to Use a Multimeter for Electronics Projects

- Oscilloscope Basics - Used Keysight Equipment
- An In-Depth Guide to Bench Power Supplies | Keysight Blogs

Intro to Permanent Prototyping

- How To Solder: A Complete Beginners Guide - Makerspaces.com
- Using Perfboard | Soldering Basics : 14 Steps (with Pictures) - Instructables
- Connector Basics - SparkFun Learn

Intro to Datasheets

- How-To: Read and Understand Technical Datasheets
- How to Read a Datasheet: Engineer's Step-by-Step Guide | MicroType

Intro to PCB design & fabrication

- PCB Design Basics: What You Need to Know Before You Start
- PCB Basics - SparkFun Learn
- Getting Started in KiCad | 10.0 | English | Documentation
    - Schematic Editor | 10.0 | English | Documentation | KiCad
    - PCB Editor | 10.0 | English | Documentation | KiCad
- PCB Component Footprint Creation in KiCad, Altium Designer, and Allegro
- **Parasitics**
    - 10 Best High-Speed PCB Routing Practices | Sierra Circuits
    - How Parasitics Impact High-Speed PCB Components

Into to PCB Schematic & Design Simulation

- Getting Started with LTspice - SparkFun Learn
- Getting Started with LTspice - Official Guide